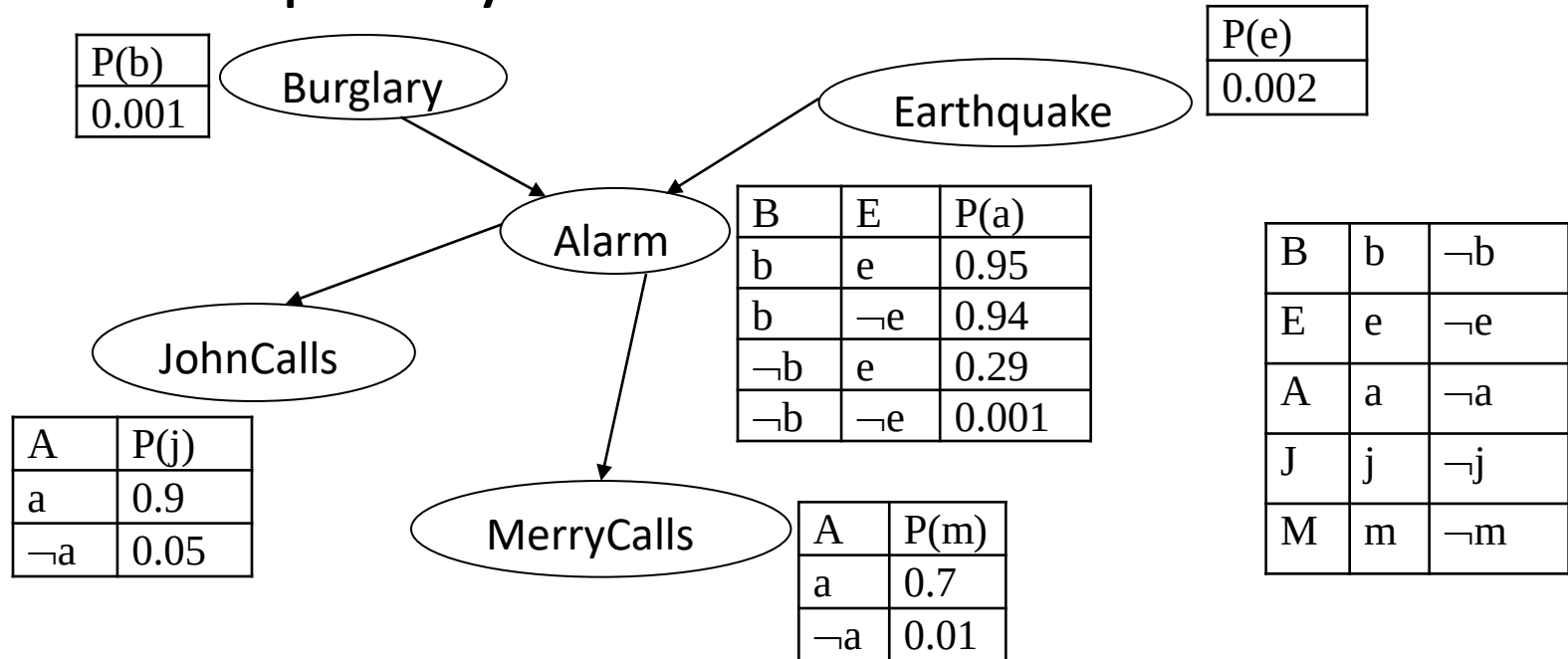


C. Another example of Bayesian Networks



➤ Making inference using the joint probabilities in the network:

$$\begin{aligned}
 \text{i) } P(a \wedge \neg j \wedge m \wedge \neg b \wedge e) &= P(a \mid \neg b \wedge e) * P(\neg j \mid a) * P(m \mid a) * P(\neg b) * P(e) \\
 &= 0.29 * 0.1 * 0.7 * 0.999 * 0.002 = \dots
 \end{aligned}$$

ii) Joint probabilities of any combination of the values of the random variables can be computed. Suppose, we need to compute

$P(b \mid a \wedge j \wedge m)$, $P(\neg b \mid a \wedge j \wedge m)$ and $P(e \mid a \wedge j \wedge m)$, $P(\neg e \mid a \wedge j \wedge m)$.

- $P(b \mid a \wedge j \wedge m) = P(b \wedge a \wedge j \wedge m) / P(a \wedge j \wedge m)$
- $P(b \wedge a \wedge j \wedge m) = P(b \wedge a \wedge j \wedge m \wedge e) + P(b \wedge a \wedge j \wedge m \wedge \neg e)$
- $P(a \wedge j \wedge m) = P(a \wedge j \wedge m \wedge b \wedge e) + P(a \wedge j \wedge m \wedge b \wedge \neg e) + P(a \wedge j \wedge m \wedge \neg b \wedge e) + P(a \wedge j \wedge m \wedge \neg b \wedge \neg e)$

➤ **Verify yourselves:**

$$P(a \wedge j \wedge m \wedge b \wedge e) = 0.0000012$$

$$P(a \wedge j \wedge m \wedge b \wedge \neg e) = 0.0005910$$

$$P(a \wedge j \wedge m \wedge \neg b \wedge e) = 0.0003650$$

$$P(a \wedge j \wedge m \wedge \neg b \wedge \neg e) = 0.0006281$$

$$\checkmark P(B \mid a \wedge j \wedge m) = \langle P(b \mid a \wedge j \wedge m), P(\neg b \mid a \wedge j \wedge m) \rangle = \langle 37\%, 63\% \rangle$$

$$\checkmark P(E \mid a \wedge j \wedge m) = \langle P(e \mid a \wedge j \wedge m), P(\neg e \mid a \wedge j \wedge m) \rangle = \langle 23\%, 77\% \rangle$$

D. Purpose of a Bayesian Network

- 1) A BN provides a complete and useful description of the domain.
- 2) It is as powerful as full joint-probability distribution.
- 3) It is significantly easier to specify.
- 4) It is human like and less costly: Only dependence of some variables from some other specific ones, not all, is considered.